



ICE-4101

**Artificial Intelligence and  
Neural Networks****Lecture-12**

**What is ANN and Artificial Neuron?  
Network Architecture, Layer, Types of Layer,  
Types of Net Archt, Feedforwad and Recurrent  
Layer**

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In this lesson, we will learn

- i. What is biological brain, nerve, and neuron
- ii. The similarity between human brain and artificial neural network or the sense of ANN
- iii. What is artificial neural network
- iv. How does ANN work
- v. Types of neural network as single-layer NN and multilayer NN
- vi. Architecture of Neural Network
- vii. Feed Forward Network and Recurrent Network
- viii. Activation function

## Human Brain



**Brain** is a **complex, nonlinear** and **parallel computer**.

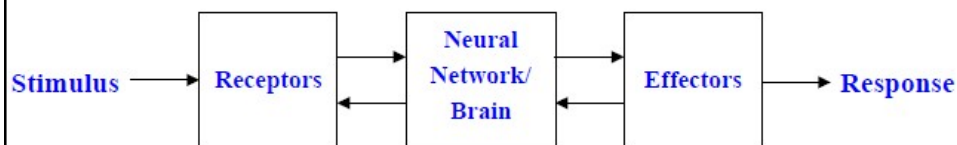
The **brain is represented by neural (nerve) net** which continually

**receives** information,  
**perceives** it, and  
**makes appropriate decisions.**

The **human nervous system** can be broken down into three stages as block diagram:

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## Human Brain

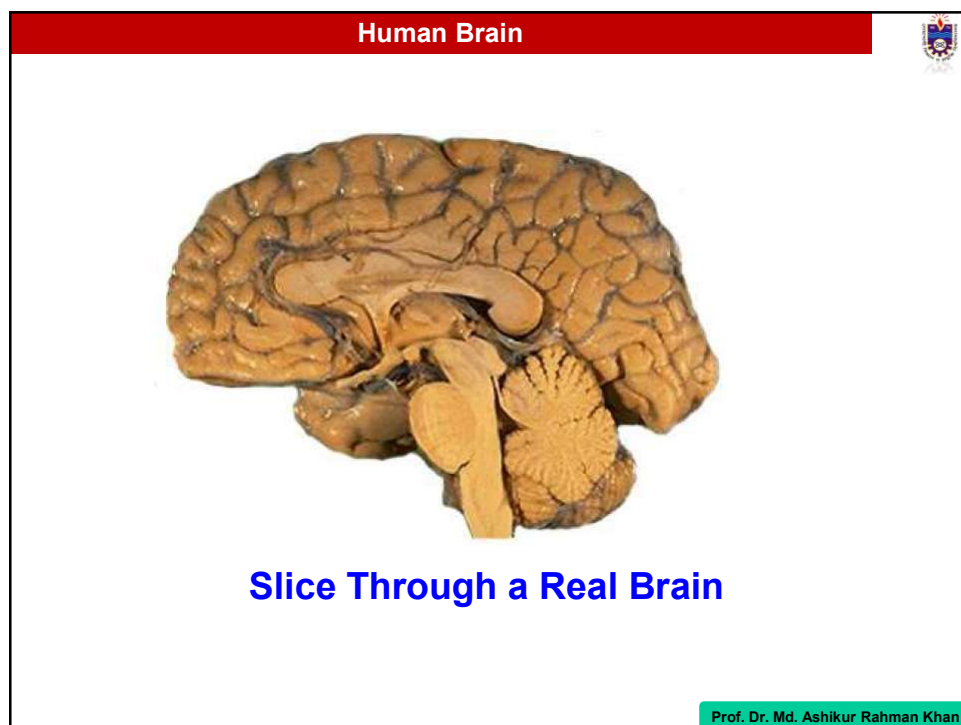
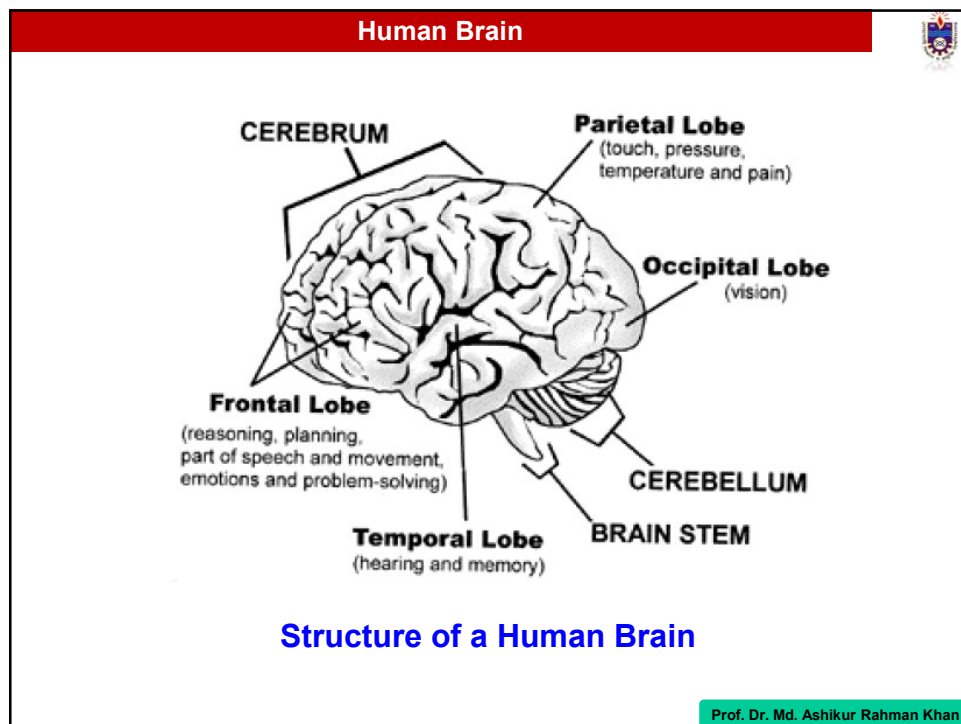


**Receptors collect information** from the environment – e.g. photons on the retina.

**Effectors convert electrical impulses** generated by the neural net into discernible (distinct visible) responses as system outputs.

The flow of information/activation is represented by arrows – feedforward (bearing signals through the system) and feedback.

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**Biological Neuron****Neuron (Biological)**

Ramon Y. Cajal **introduced the idea of neurons**

The **basic of biological neural systems are nerve cells (Neuron).**

There are approximately **10 billion neurons in the human cortex.**

**Each biological neuron is connected to several thousands (on average 10,000) of other neurons**

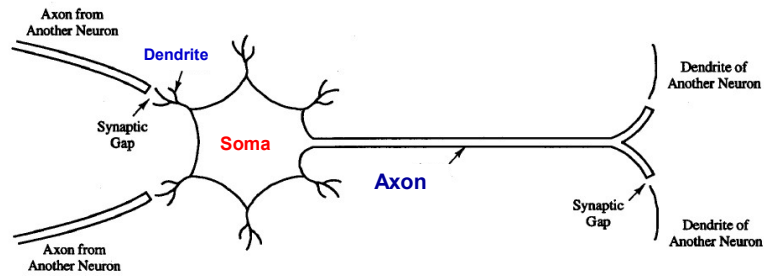
similar to the connectivity in powerful parallel computers.

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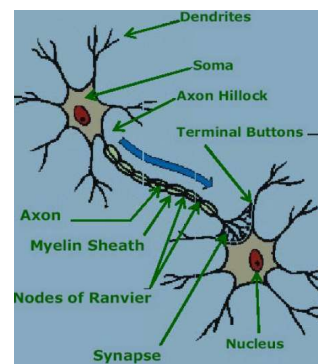
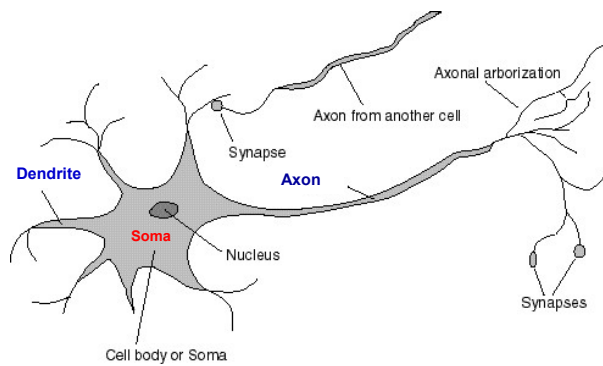
**Biological neuron is composed of**

- **Dendrites,**
- **Soma (cell body), and**
- **Axon**

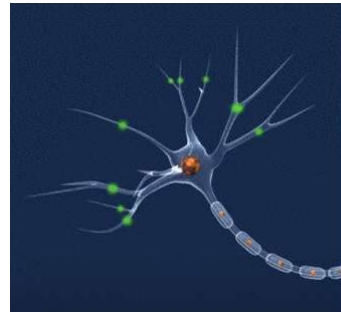
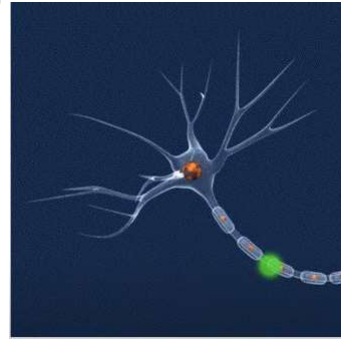
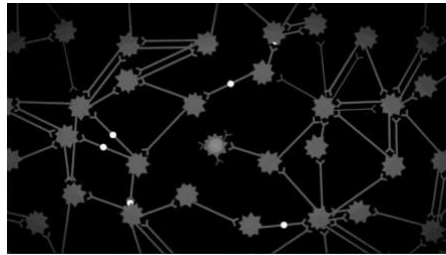
## Biological Neuron



## Biological Neuron



Transmit signal from one neuron to another neuron



H:\Course No. ICE-4101- Neural Network\Lectures Final on March 2019 (4th Batch)\Nervous Tissue

## Biological Neuron



**i. Dendrites** receive signals (electrical impulses) from other neurons.

These signals are transmitted across a synaptic gap by means of a chemical process and scaled.

The dendrites behave as input channels.

**ii. Soma** is the cell body (**nucleus**) which sums the incoming signals.

The neuron's cell body (soma) **processes the incoming activations and converts them into output activations.**

## Biological Neuron



When the soma has gathered enough information (input/signal), the cell transmitted a signal over its axon to other cells. The action of transmitting a signal **is known as a cell fire**.

Neurons are massively interconnected.

**The junctions that allow signal transmission between the axons and dendrites are called **synapses**.**

**iii. Axons** are fibres acting as transmission lines that send activation to other neurons.

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## Artificial neural network



**The characteristics of biological neural networks that serve as the inspiration for artificial neural networks.**

An **artificial neural network** is an information-processing system that has certain performance characteristics in common with biological neural networks.

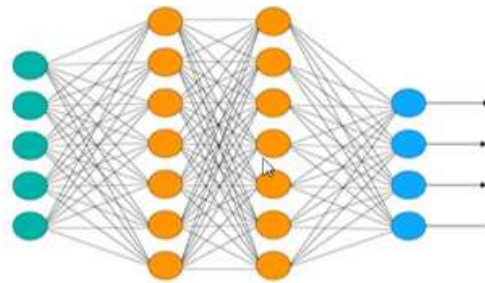
An **artificial neural network is defined** as “a data processing system consisting a large number of simple, highly interconnected processing elements (artificial neurons) in an architecture similar to the structure of the cerebral cortex of the brain”.

## Neural Network

A neural Network is a computing system that is based on the biological neural network that make up the human brain.



A neural network is made of a collection of units or nodes called neurons. These neurons are connected one to the other by means of a connection called synapse.



## Artificial neural network



**Artificial neural networks have been developed** as generalizations of mathematical models of human cognition or neural biology, **based on the assumptions** that:

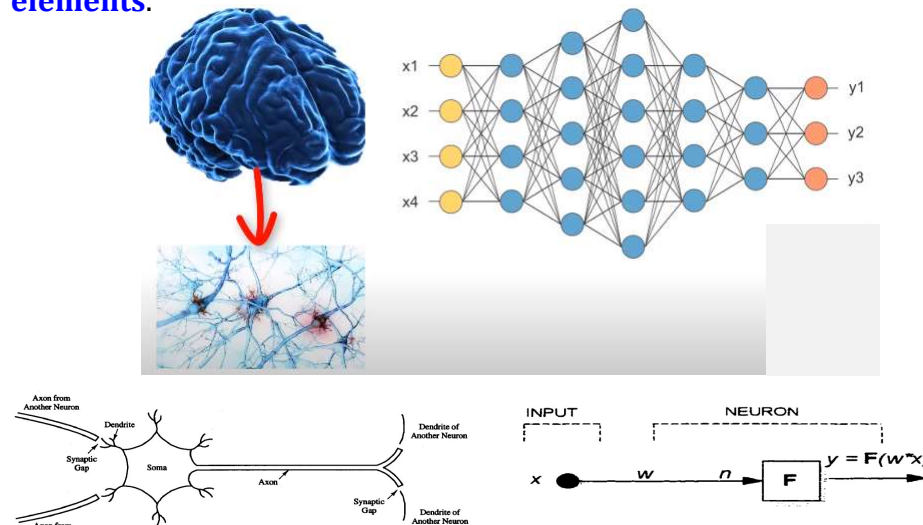
- i. Information **processing occurs** at many simple elements called **neurons (processing element)**.
- ii. Signals are **passed between** neurons over connection **links**.
- iii. Each connection link has an **associated weight**, which, in a typical neural net, multiplies the signal transmitted.
- iv. Each neuron applies an **activation function** (usually nonlinear) to its net input (sum of weighted input signals) to determine its output signal.

| Brain                            |                     |                  |
|----------------------------------|---------------------|------------------|
| Comparison of Brain and computer |                     |                  |
|                                  | <i>Human</i>        | <i>Computer</i>  |
| <i>Processing Elements</i>       | 100 Billion neurons | 10 Million gates |
| <i>Interconnects</i>             | 1000 per neuron     | A few            |
| <i>Cycles per sec</i>            | 1000                | 500 Million      |
| <i>2X improvement</i>            | 200,000 Years       | 2 Years          |

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### Artificial neural network

Both biological and artificial neural networks have the **processing elements**.



**Artificial and Biological Neurons Scheme**

## Artificial Neuron

## Artificial Neuron



A **neural net** consists of a large number of simple processing elements called **neurons, units, cells, or nodes**.

The “core” element of a neural network is the neuron (processing element).

Neurons are connected to each other with a set of links, called synapses and each synapse is described by a **synaptic weight**.

**Transfer (activation) functions** (usually non-linear) are applied by each neuron to its net input to compute its (neuron's) output.

## Artificial Neuron



## Artificial Neuron

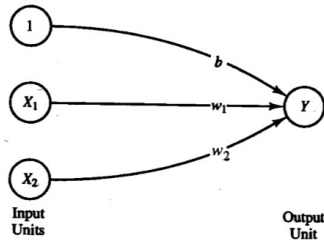
A **single input  $x$**  is transmitted through a connection that **multiplies its strength by a weight  $w$**  to form a weighted input,  $wx$  which is used as the only argument to the **transfer function** (or **activation function**) that generate a product  $y$ .

The **weight and bias** are parameters that can be adjusted so that the network can exhibit some desired behavior during its training phase.

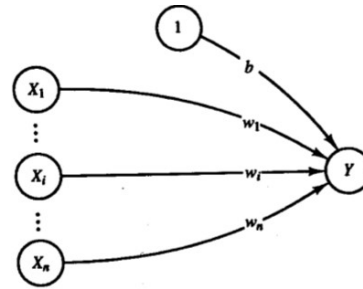
The **Bias,  $b$** , is much like a weight, except that it has a constant input of 1.

## Bias and Thresholds

A bias acts exactly as a weight on a connection from a unit whose activation is always 1. Increasing the bias increases the net input to the unit.



Single-layer neural network for logic functions



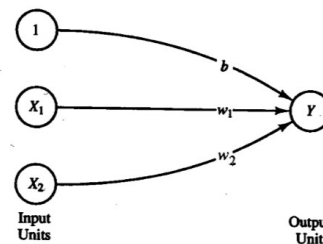
Perceptron to perform single classification

## Bias and Thresholds

If a bias is included, the activation function is typically taken to be

$$f(\text{net}) = \begin{cases} 1 & \text{if net} \geq 0; \\ -1 & \text{if net} < 0; \end{cases}$$

where  $\text{net} = b + \sum_i x_i w_i.$



Some authors do not use a bias weight, but instead of bias use a fixed threshold for the activation function. In that case

$$f(\text{net}) = \begin{cases} 1 & \text{if net} \geq \theta; \\ -1 & \text{if net} < \theta; \end{cases}$$

where

$$\text{net} = \sum_i x_i w_i.$$

## Role of Bias and Threshold

1. The boundary between the value of  $x_1$  and  $x_2$  for which the network gives a positive response and the values for which it gives a negative response is the **separating line**

$$b + x_1w_1 + x_2w_2 = 0$$

assuming that  $w_2 \neq 0$

$$x_2 = -\frac{w_1}{w_2}x_1 - \frac{b}{w_2}$$

The requirement for a positive response from the output unit is that the net input it receives, namely,

$$b + x_1w_1 + x_2w_2 > 0$$

## Role of Bias and Threshold

2. If one thinks in terms of a threshold, the requirement for a positive response from the output unit is that the net input it receives, namely

$$x_1w_1 + x_2w_2 > \textit{threshold}.$$

This gives the equation of line separating positive from negative output as

$$x_1w_1 + x_2w_2 = \theta$$

assuming that  $w_2 \neq 0$

$$x_2 = -\frac{w_1}{w_2}x_1 + \frac{\theta}{w_2}$$

3. In some problem (like perceptron nn), both bias and threshold are considered as

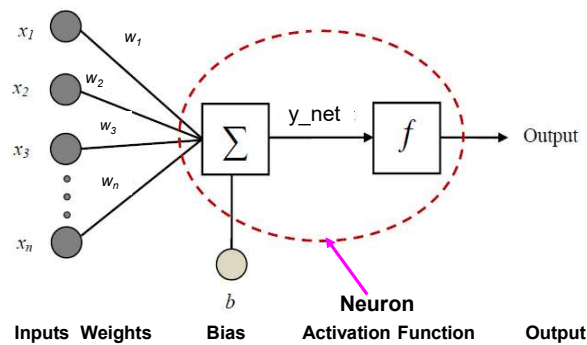
The line separating the region of positive response from the region of zero response is as follow

$$w_1x_1 + w_2x_2 + b > \theta$$

The line separating the region of zero response from the region of negative response is as follow

$$w_1x_1 + w_2x_2 + b < -\theta$$

### A Multiple Input Neuron Model



A Multiple Input Neuron Model

$$y_{net} = ?$$

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## Artificial Neural Network



The net input to the transfer function ( $f$ ) is defined as the sum of the bias and the weighted inputs, such that:

$$y\_net = \sum_{i=1}^n w_i x_i + b$$

where  $n$  = total number of input variables

wx = product of input and weight

x = Inputs

w = weights

The output for the neuron is computed using the [transfer function](#), such that

$$output = f(y\_net)$$

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$$output = f(y\_net)$$

$$or output = f(net\_input)$$

$$or y = f(net)$$

$$or y = f(I)$$

$$or y = \phi(I)$$

## A simple Neuron

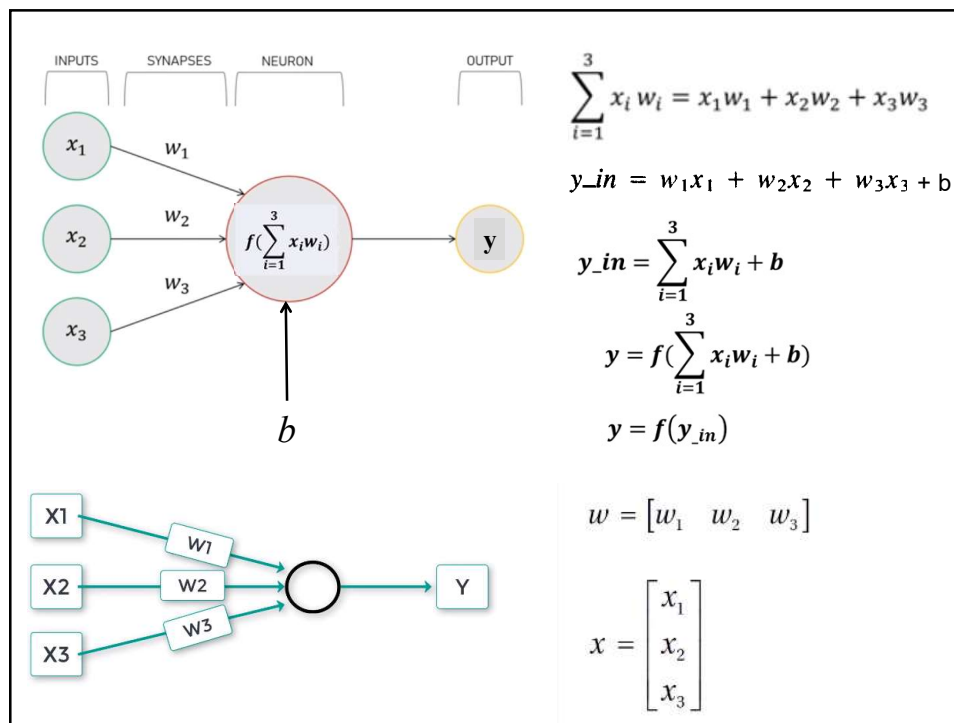
**For example:** Consider a neuron Y, illustrated in following figure.

It receives inputs from neurons  $x_1$ ,  $x_2$ , and  $x_3$ .

The activations (output signals) of these neurons are  $x_1$ ,  $x_2$ , and  $x_3$  respectively.

The weights on the connections from  $x_1$ ,  $x_2$ , and  $x_3$  to neuron y are  $w_1$ ,  $w_2$ , and  $w_3$ , respectively.

The net input,  $y_{in}$ , to neuron y is the sum of the weighted signals from neurons  $x_1$ ,  $x_2$ , and  $x_3$ , i.e.



## Layer

Neurons are arranged in layers.

A layer is the combination of the

- weights,
- net input,
- bias, and
- transfer function.

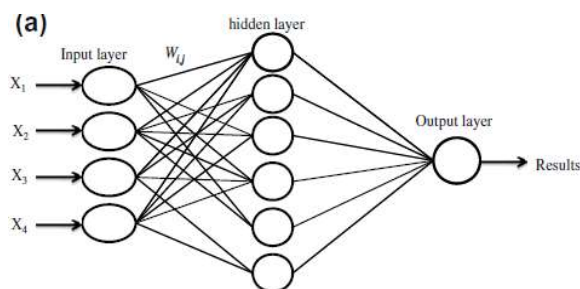
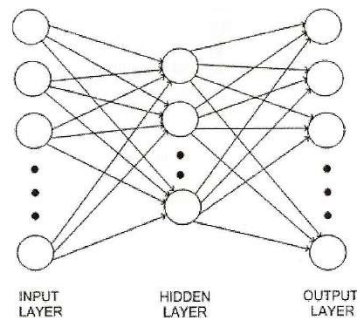
A **hidden unit** is that unit which is neither an input unit nor an output unit.

A network may have many or no hidden layers, and their role is to improve the network's performance.

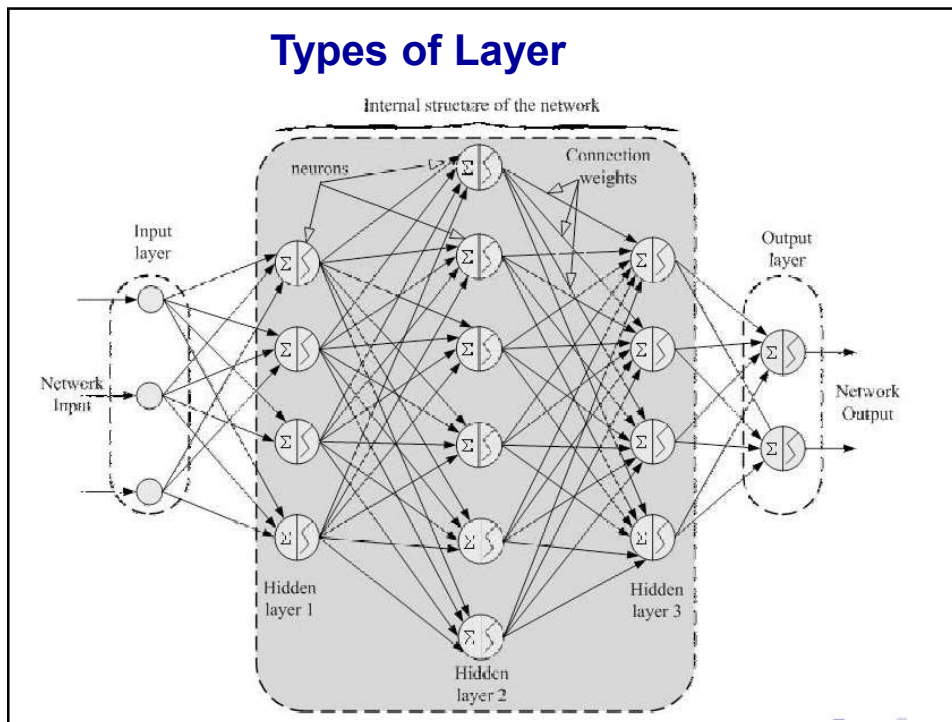
## Types of Layer

Three Types of Layers:

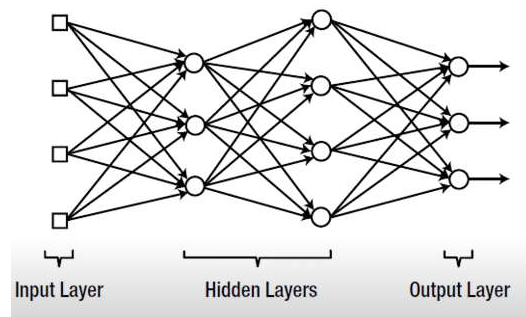
- i. Input layer
- ii. Hidden layer
- iii. Output layer



## Types of Layer



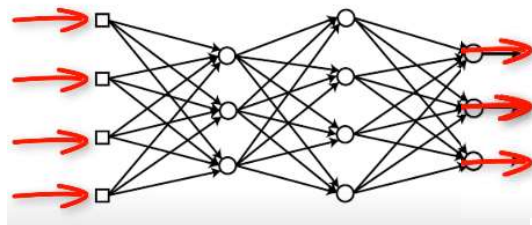
## Description of Neural Network



- Inputs nodes transmit the input signals only.
- They do not calculate the weighted sum and do not use activation function that's why **the input neurons are not counted as a layer.**
- Output nodes give output.

## Description of Neural Network

- A **hidden unit** is that unit which is neither an input unit nor an output unit.
- The layers between input and output nodes are called hidden layers. They are called hidden layer because they are not accessible from the outside of NN.
- **A network may have many or no hidden layers**, and their role is to improve the network's performance.



In layered neural network, the signals enter the input layer, pass through the hidden layers and leave through output layer.

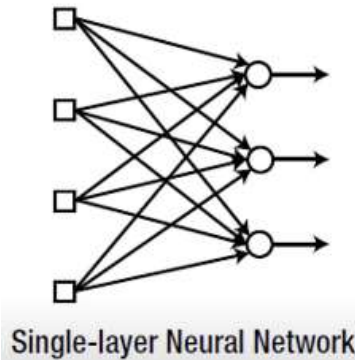
The input signals advance layer by layer.

The nodes of a layer receive signals, process them and transmit them to the nodes of the next layer.

## Simple Layer and Complex layer NN

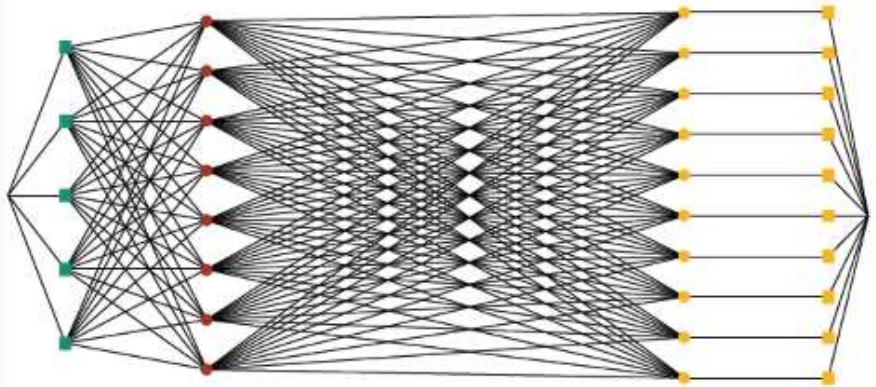
### Simple Layer

(Single-layer Neural Network)



This type of NN has only input and output nodes.

### More Complex Neural Network



The NN may be from simple architecture to more and more complex architecture.

## Classification

Neural nets are often classified as

- Single layer network
- Multilayer network.

### Single layer network

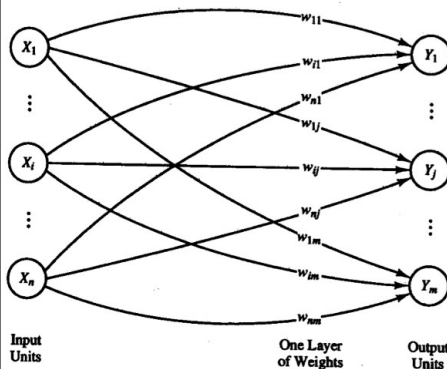


Figure 1.4: A single-layer neural network

### Multilayer network.

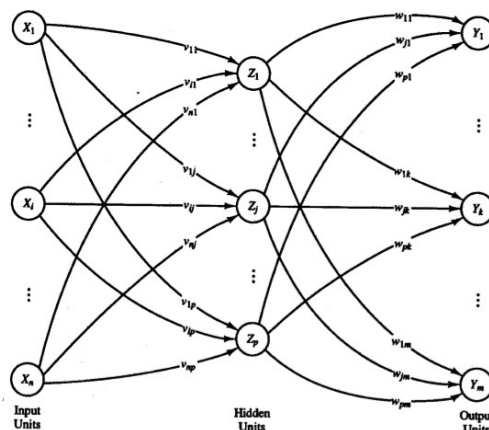
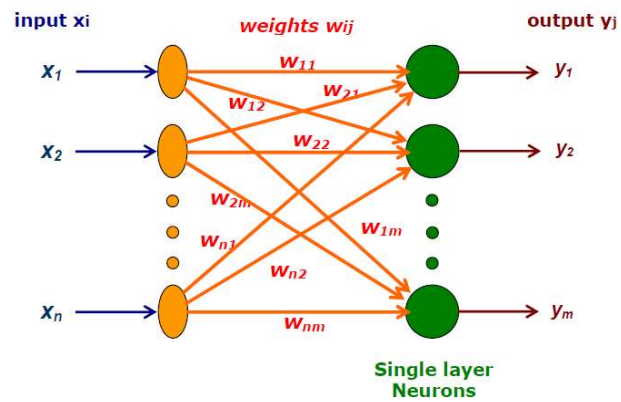


Figure 1.5: A multilayer-layer neural network

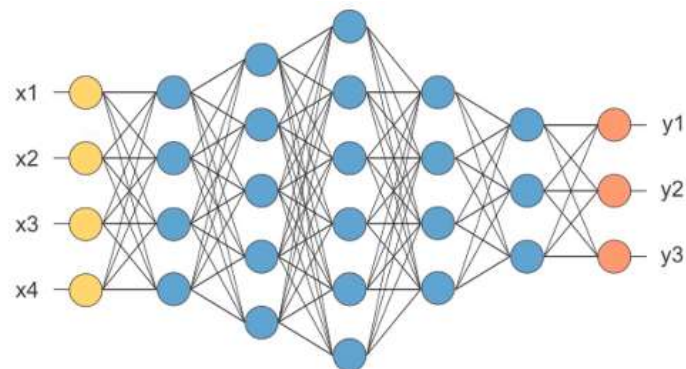


## Single-layer Network



A single layer neural network

## Multilayer Network



Multilayer nets can solve more complicated problems than can single-layer nets, but training may be more difficult.



## Multilayer network

### A multilayer network

A multilayer network (net) is a net with one or more layers (or levels) of nodes (also called **hidden units**) **between the input units and output units**. In general, **multiple networks contain one or more hidden layers**. The **hidden layers follow the input layer**.

Typically, there is a layer of weights between two adjacent levels of units (input, hidden, or output)

Two or more neurons can be connected to a layer or in multiple layers. The MLP neural network is formed from numerous neurons with parallel connections, which are joined in several layers.

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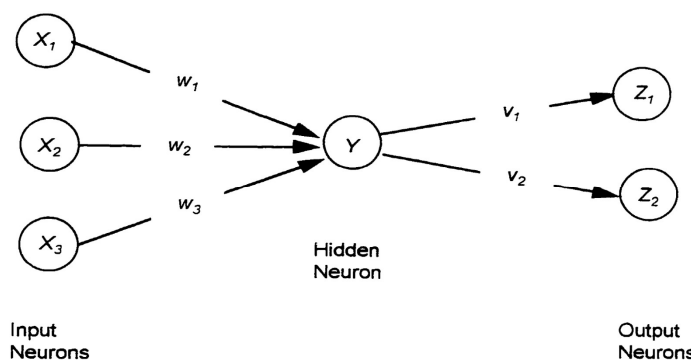
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## Multilayer network

Multilayer nets can solve more complicated problems than can single-layer nets, **but training may be more difficult**.

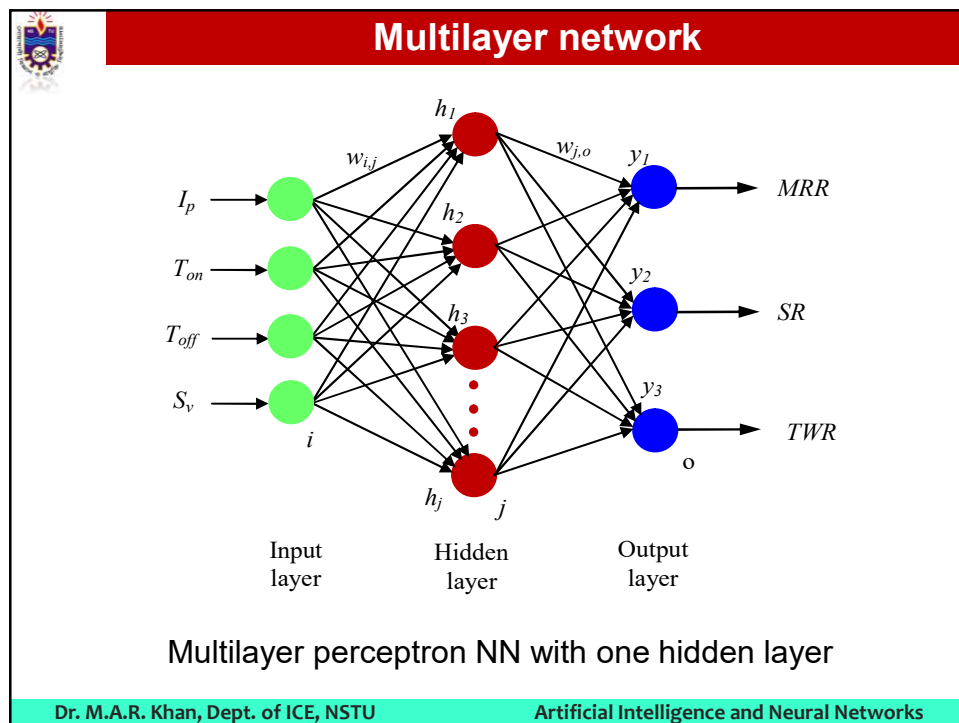
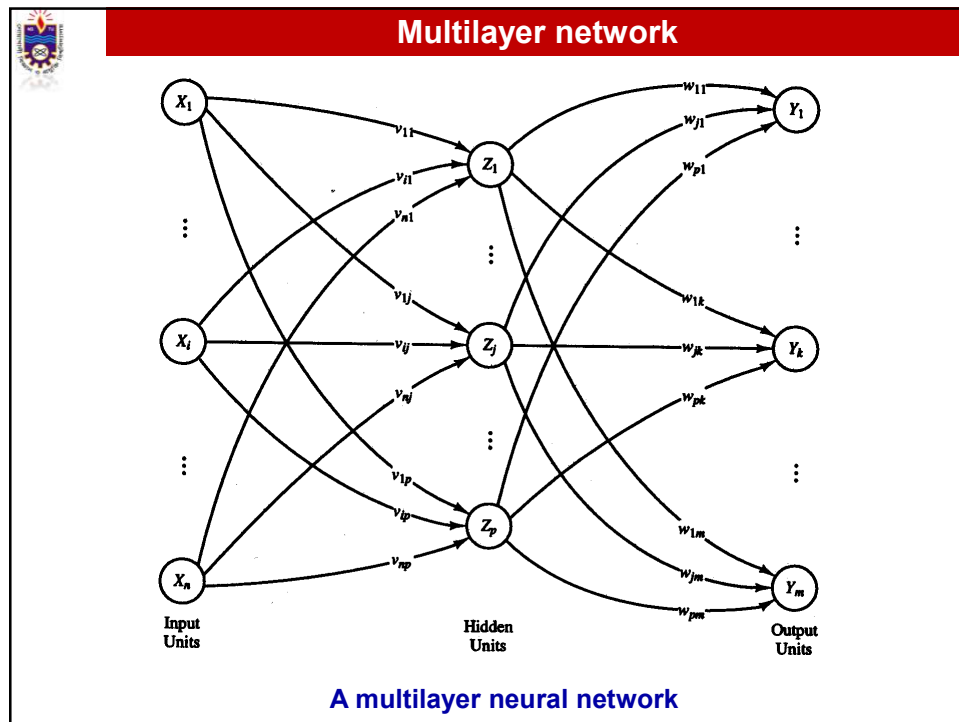
However, in some cases, training may be more successful, because it is possible to solve a problem that a single-layer network cannot be trained to perform correctly at all.

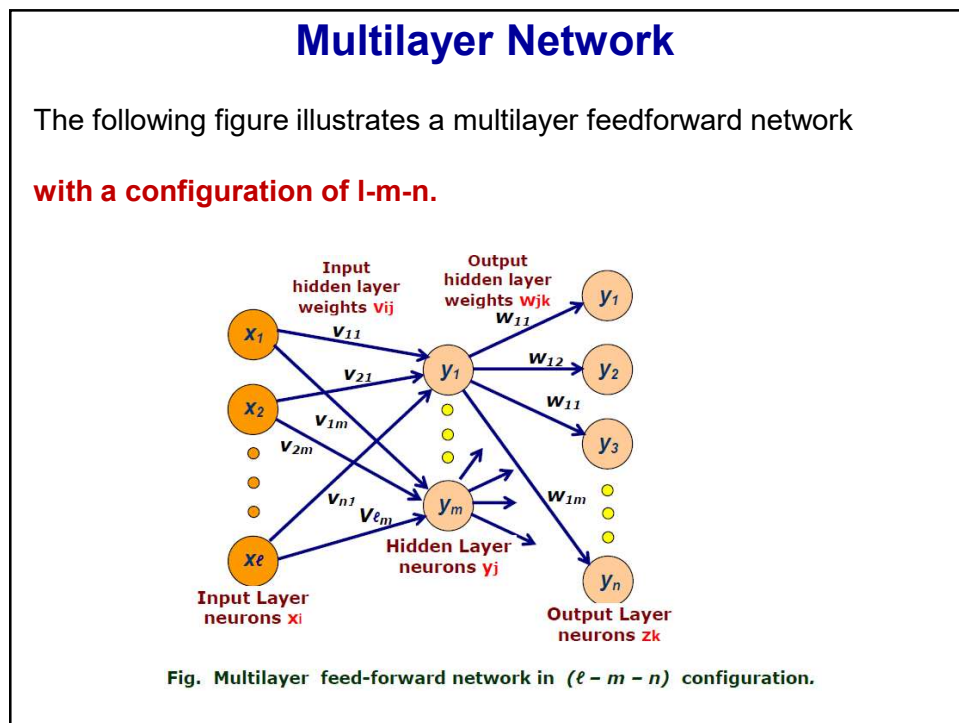
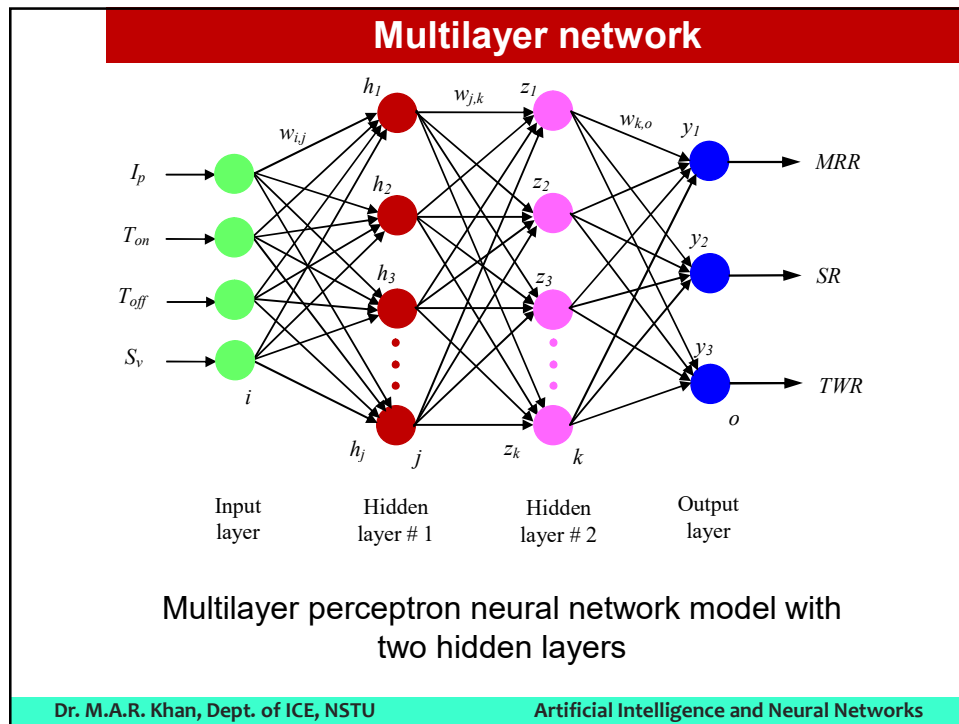


### A two layer neural network

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## Multilayer Network

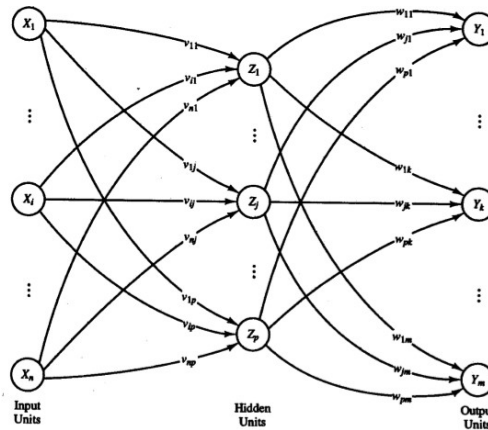


Figure 1.5: A multilayer-layer neural network

## Neural Network Characteristics



### NN characteristics

A neural network is characterized by:

- (i) Its pattern of connections between the neurons (called its **architecture**)
- (ii) Its method of determining the weights on the connections (called its **training, or learning, algorithm**), and
- (i) Its **activation function** (also called **transfer function**).

## Network Architecture

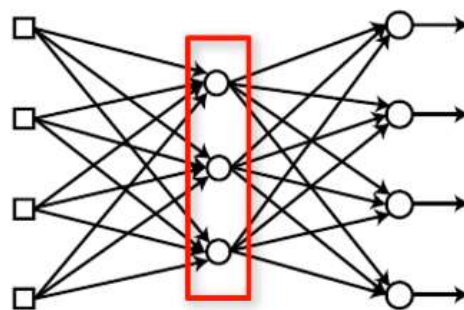
The arrangement of neurons into layers and the connection patterns within and between layers is called the net architecture.

Many neural nets have an **input layer** in which the activation of each unit is equal to an external input signal.

The net illustrated in following figure consists of input units, output units, and one hidden unit.

## Multilayer Network

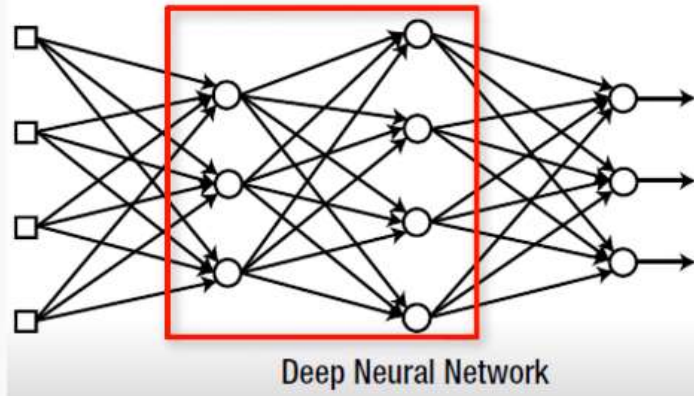
### Shallow Neural Network



(Shallow) Multi-layer Neural Network

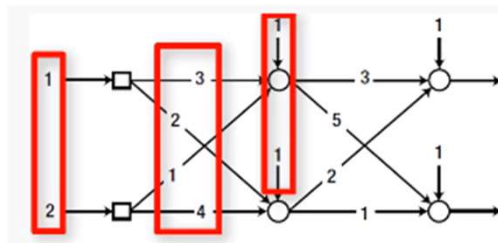
When a hidden layer is added to single layer NN is called multilayer or Shallow NN.

## Deep Neural Network



When there are two or more than two hidden layer is called Deep NN

## Example



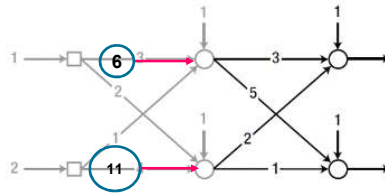
$$\text{Weighted sum: } v = (3 \times 1) + (1 \times 2) + 1 = 6$$

$$\text{Weighted sum: } v = (2 \times 1) + (4 \times 2) + 1 = 11$$

$$v = \begin{bmatrix} 3 \times 1 + 1 \times 2 + 1 \\ 2 \times 1 + 4 \times 2 + 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}$$

$$v = Wx + b$$

## Example (Contd.)



$$\text{Weighted sum: } v = \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 41 \\ 42 \end{bmatrix}$$

$$\text{Output: } y = \varphi(v) = v = \begin{bmatrix} 41 \\ 42 \end{bmatrix}$$

## Example:

A neuron consists of four inputs with the weights as shown in below.

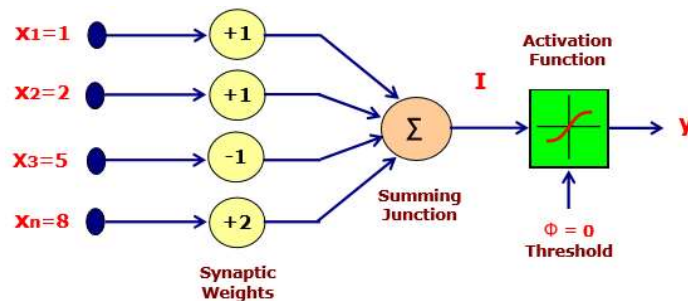


Fig Neuron Structure of Example

**Solution:**

The output **I** of the network, prior to the activation function stage, is

$$\mathbf{I} = \mathbf{X}^T \cdot \mathbf{W} = \begin{bmatrix} 1 & 2 & 5 & 8 \end{bmatrix} \bullet \begin{bmatrix} +1 \\ +1 \\ -1 \\ +2 \end{bmatrix} = 14$$

$$= (1 \times 1) + (2 \times 1) + (5 \times -1) + (8 \times 2) = 14$$

With a binary activation function the outputs of the neuron is:

$$\mathbf{y} \text{ (threshold)} = 1;$$

## Types of Network Architecture

There are several classes of **neural network architecture**.

ANN are classified according to the following:

- **Feed-forward Network**
- **Recurrent Network**

## Feed-forward Network

**Feedforward nets** are those nets in which the signals flow from the input units to the output units, in a forward direction.

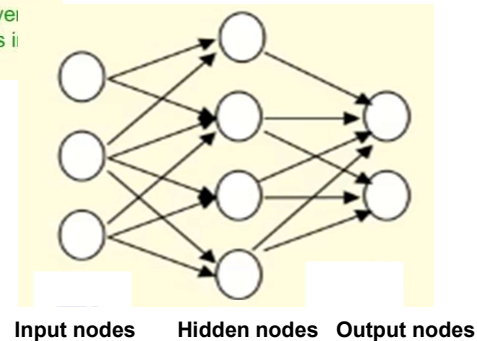
The synaptic links carrying weights connect every input neuron to the output neuron but not vice-versa.

Such a network is said to be **feed-forward**.

The single-layer and multilayer nets illustrated in Figures 1.4 and 1.5 are examples of feedforward nets.

## Feed-forward neural networks

- These are the commonest type of neural network in practical applications.
  - The first layer is the input and the last layer is the output.
  - If there is more than one hidden layer, we call them “deep” neural networks.
- They compute a series of transformations that change the similarities between cases.
  - The activities of the neurons in each layer are a non-linear function of the activities in the layer below.



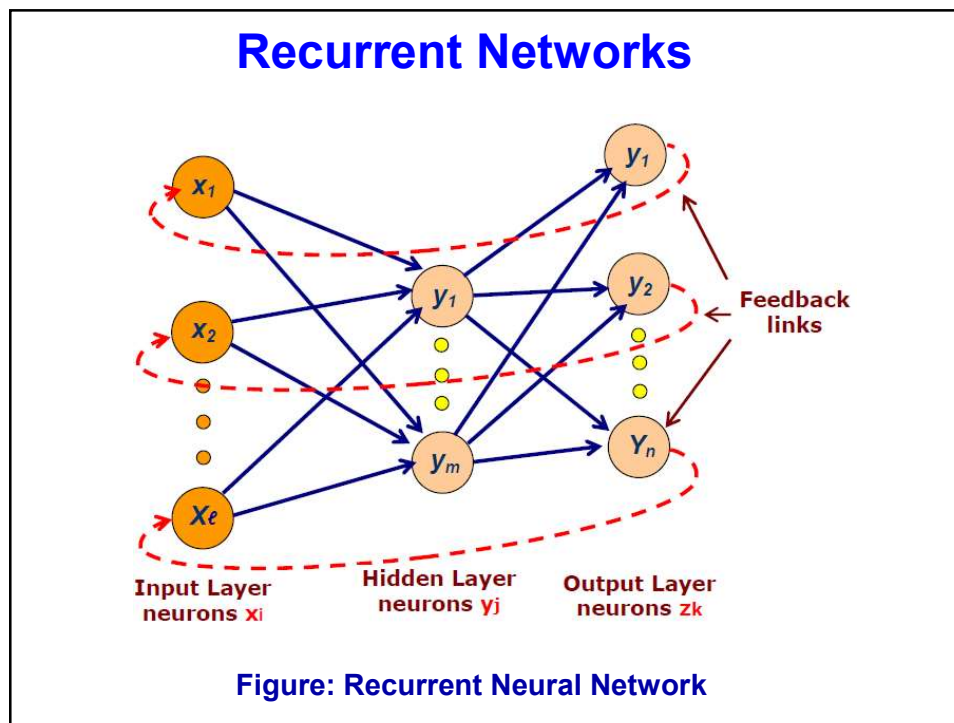
## 2. Recurrent Networks

These networks differ from feed-forward network architectures in this sense that there is at least one feed back loop.

Thus, in these networks, there could exist one layer with feedback connections as shown in following figure.

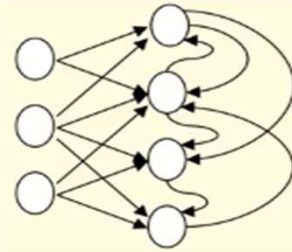
There could also be neurons with self-feedback, i.e. the output of a neuron is fed back into itself as input.

The fully interconnected competitive net in Figure 1.6 is an example of a recurrent net, in which there are closed-loop signal paths from a unit back to itself.



## Recurrent networks

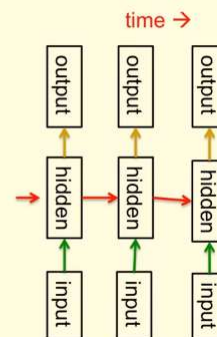
- These have directed cycles in their connection graph.
  - That means you can sometimes get back to where you started by following the arrows.
- They can have complicated dynamics and this can make them very difficult to train.
  - There is a lot of interest at present in finding efficient ways of training recurrent nets.
- They are more biologically realistic.



Recurrent nets with multiple hidden layers are just a special case that has some of the hidden→hidden connections missing.

## Recurrent neural networks for modeling sequences

- Recurrent neural networks are a very natural way to model sequential data:
  - They are equivalent to very deep nets with one hidden layer per time slice.
  - Except that they use the same weights at every time slice and they get input at every time slice.
- They have the ability to remember information in their hidden state for a long time.
  - But its very hard to train them to use this potential.



### An example of what recurrent neural nets can now do (to whet your interest!)

- Ilya Sutskever (2011) trained a special type of recurrent neural net to predict the next character in a sequence.
- After training for a long time on a string of half a billion characters from English Wikipedia, he got it to generate new text.
  - It generates by predicting the probability distribution for the next character and then sampling a character from this distribution.

### Activation function



The basic operation of an artificial neuron involves summing its weighted input signal and applying an output, or **activation function**.

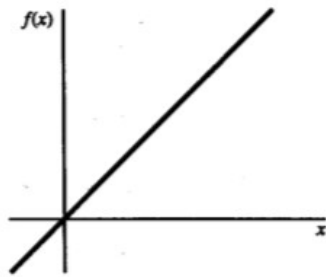
Typically, the **same activation function is used for all neurons in any particular layer of a neural net**, although this is not required.

**In most cases, a nonlinear activation function is used.** In order to achieve the advantages of multilayer nets, compared with the limited capabilities of single-layer nets, **nonlinear function are required**.

## Activation function

### i. Identity Function

It is Linear function



$$f(x) = x \text{ for all } x$$

It is also expressed as

$$y = \text{purelin}(x)$$

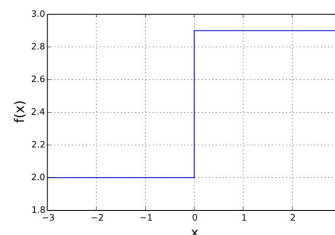
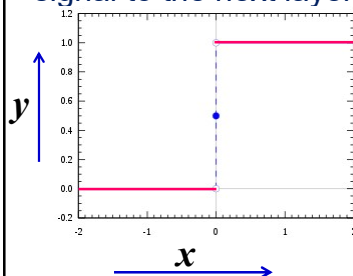
Figure 1.7 Identity function.

Usually, Non-linear functions are used in Neural network.

These are as follows:

### ii. Hard Limit function/Binary step function/ threshold or Heaviside function

A binary step function is a threshold-based activation function. If the input value is above or below a certain threshold, the neuron is activated and sends exactly the same signal to the next layer.



$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$

In step function the output is a

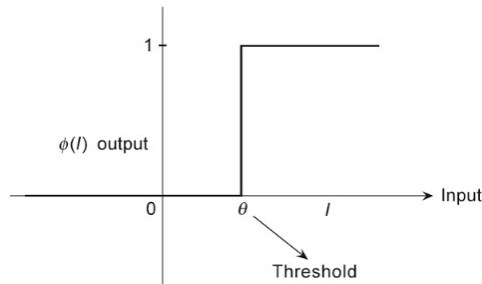
**Binary 1 or 0**

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Or **Bipolar 1 or -1**

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

If a threshold is used the step function is called threshold or Heaviside function



Binary step function (with threshold  $\theta$ ):

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

A binary step function is generally used in the Perceptron learning rule for classifier. It thresholds the input values to 1 and 0, if they are greater or less than zero, respectively.

In Matlab, it is also expressed as

**y = hardlim(x)**

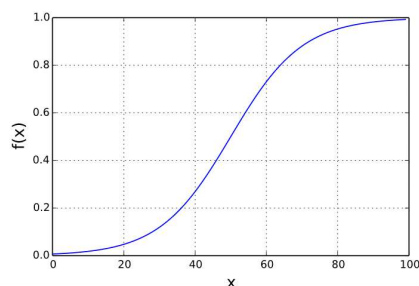
### iii. Sigmoid function or logistic function



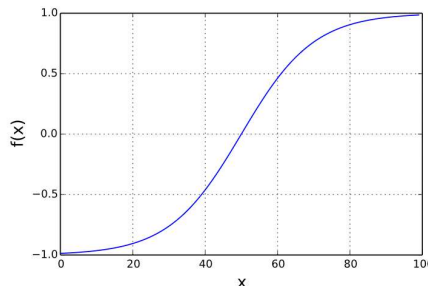
The sigmoid or **Log-Sigmoid** or logistic function maps the input values in the range (0,1), which is essentially their probability of belonging to a class. So, it is mostly used for multi-class classification.

It also has a low convergence rate.

The sigmoid function can be scaled to have any range of output values, depending upon the problem.



**Binary Sigmoid function**

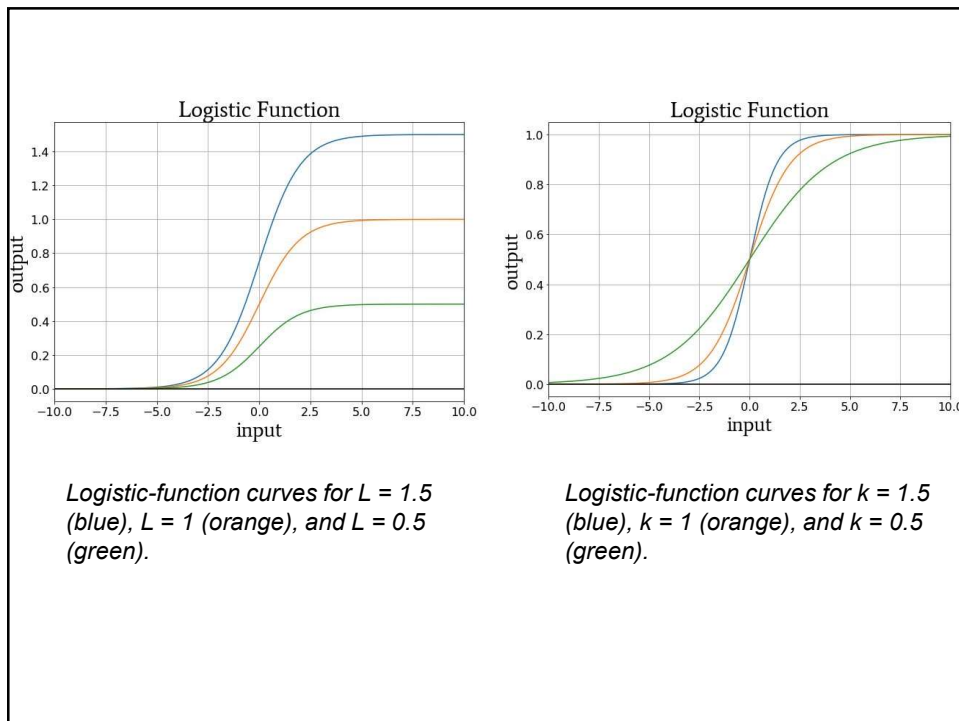


**Bipolar Sigmoid function**

The adjective “sigmoid” refers to something that is curved in two directions. There are various sigmoid functions, and we’re only interested in one. It’s called the logistic function, and the mathematical expression is fairly straightforward:

$$f(x) = \frac{L}{1 + \exp(-kx)}$$

The constant L determines the curve’s maximum value, and the constant k influences the steepness of the transition. The plot below shows examples of the logistic function for different values of L, and the following plot shows curves for different values of k.



## Sigmoid function (Contd.)

A **logistic function** or **logistic curve** is a common S-shaped curve with equation

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}},$$

where

$x_0$  = the  $x$  value of the sigmoid's midpoint,

$L$  = the curve's maximum value,

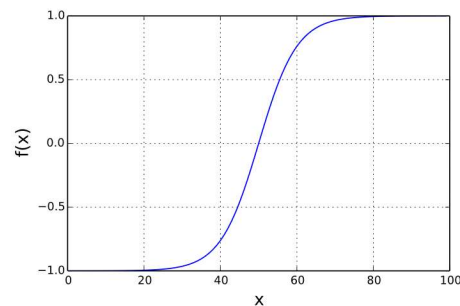
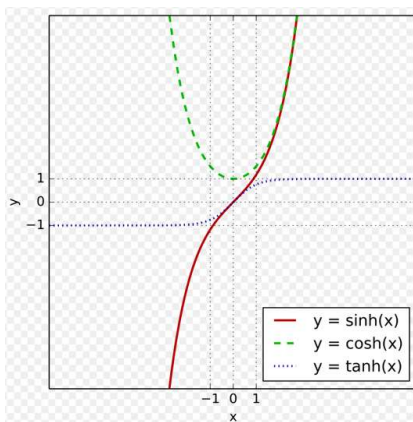
$k$  = the logistic growth rate or steepness of the curve.

The **standard logistic function** is the logistic function with parameters  $k = 1$ ,  $x = 0$ ,  $L = 1$ , which yields

$$f(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1} = \frac{1}{2} + \frac{1}{2} \tanh\left(\frac{x}{2}\right).$$

$$f(x) = \frac{1}{1 + e^{-x}}$$

## iv. Hyperbolic Tangent functions (tanh)



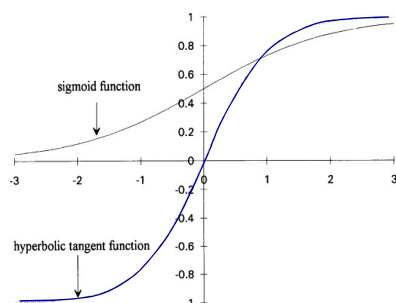
**Range -1 to 1**

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$f(x) = \tanh(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$$

Like the Sigmoid function

#### iv. Hyperbolic Tangent functions (tanh)

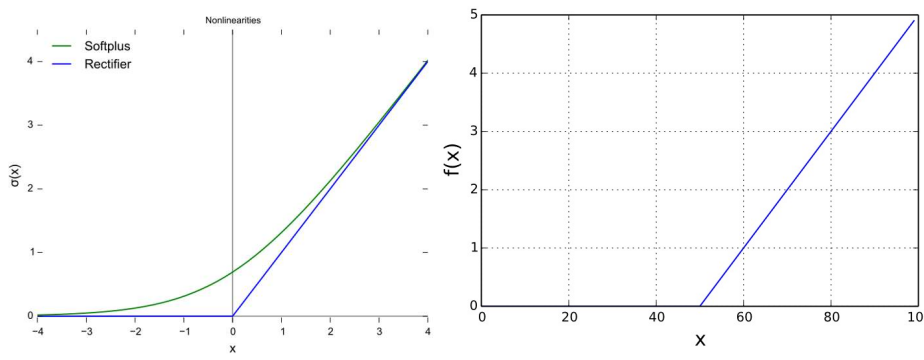


- the slope of the hyperbolic tangent function is much greater than the slope of the sigmoid function;
- the hyperbolic tangent function has a negative response for a negative input value and a positive response for a positive input value, while the sigmoid function always has a positive response.

#### v. ReLU (Rectified Linear Unit)

A rectified linear unit has **the output 0 if its input is less than or equal to 0**, otherwise, **its output is equal to its input**. It has been widely used in **convolutional neural networks**.

It is also superior to the sigmoid and tanh activation function. It allows for faster and effective training of **deep neural architectures**.



the most popular activation function for deep neural networks.

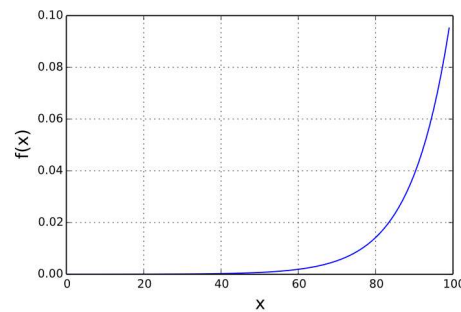
Range 0 to  $\infty$

$$f(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ x & \text{for } x > 0 \end{cases}$$

## vi. Softmax

The softmax function gives the probabilities that any of the classes are true. It produces values in the range (0,1).

This function is widely used in multiple classification logistic regression models.



$$\sigma(z)_j = \frac{e^{z_j}}{\sum_{k=1}^k e^{z_k}} \text{ for } j = 1 \dots k.$$

## Advantages of neural network



**What are the advantages of neural networks over conventional computers?**

Neural networks learn by example.

They are more fault tolerant because they are always able to respond and small changes in input do not normally cause a change in output.

Because of their parallel architecture, high computational rates are achieved.



# THANKS

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